Evolution vs. Creationism in the Classroom: The Lasting Effects of Science Education

Benjamin W. Arold

The Quarterly Journal of Economics, 2024

Prepared for ifo Summer Course 2026, by Benjamin W. Arold

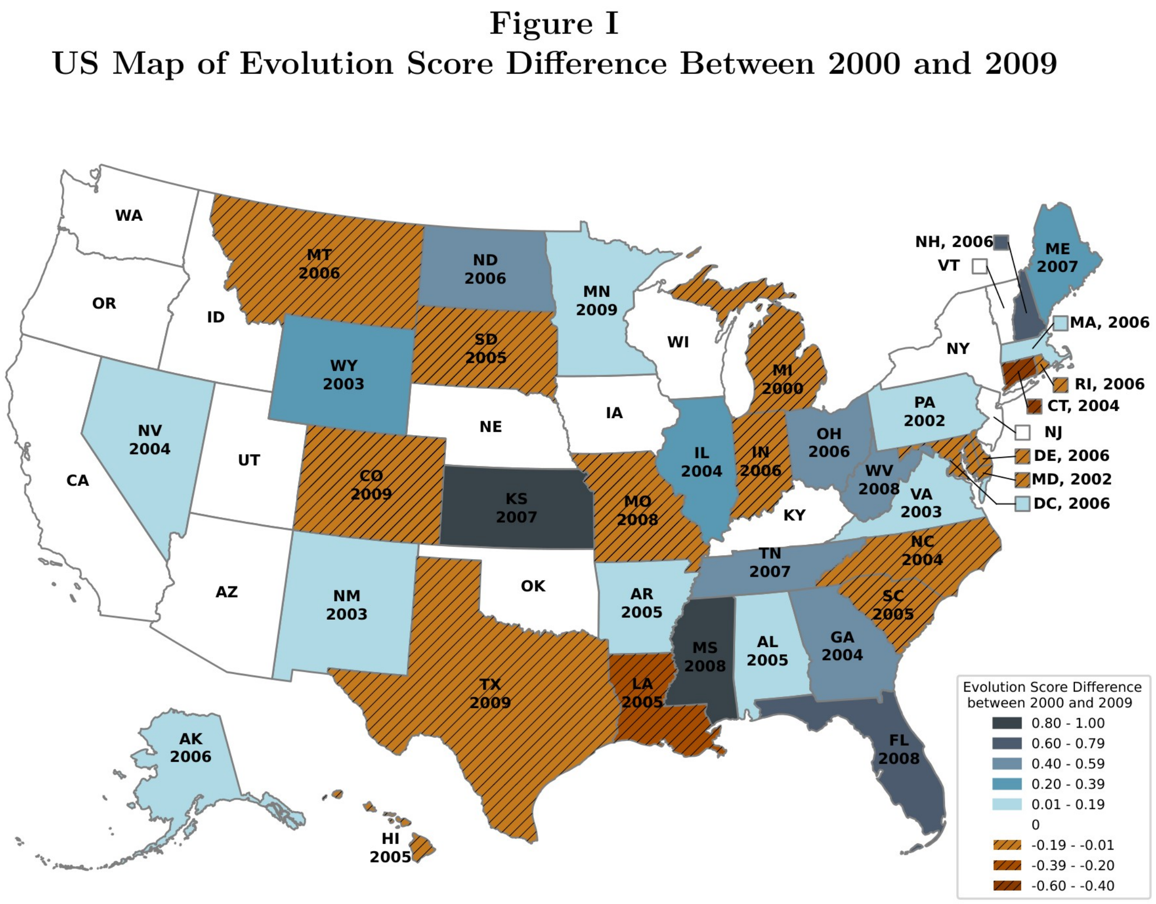
Motivation & Research Question

- Anti-scientific attitudes impose large costs on public health (COVID misinformation), the environment (climate denial), and the economy.
- Evidence on the DETERMINANTS of such attitudes is scant -- yet crucial for effective policy.
- This paper isolates the CONTENT of high-school science education as one policy-relevant determinant.
  - Focus on evolution theory: fundamental to the life sciences, yet highly contested (98% of scientists vs. only 65% of the US public accept human evolution).
- Question: does exposure to the teaching of evolution causally shape (i) evolution KNOWLEDGE, (ii) evolution BELIEF in adulthood, and (iii) the choice to WORK IN LIFE SCIENCES?

Setting: US Science Standards & Evolution Reforms

- US State Science Education Standards define what students must master; they drive local curricula, textbook content, and standardized testing.
- 88% of high-school biology teachers say they teach evolution to meet the Science Standards.
- 2000-2009: 15 states REDUCED and 22 states EXPANDED evolution coverage; 13 unchanged.
- Reforms are decided by State Boards of Education (appointed or elected), after drafting committees and public hearings.
  - Whether a reform happens is not random -- but its exact YEAR is as-good-as-random, driven by institutional idiosyncrasies (election cycles, board tenure up to 9 years, drafting delays).
- Implementation is swift: textbooks, lesson plans, and testing adjust within ~1-3 years; teacher hiring unaffected -> effects are lower bounds.

Measuring Evolution Coverage: The Evolution Score



Note: Figure I: US map of the evolution-score difference (2009 minus 2000). Blue = increase, orange stripes = decrease, white = no change; year labels mark reform years. Identification comes from within-state changes. Source: Lerner (2000), Mead & Mates (2009).

- Evolution score (Lerner 2000; Mead & Mates 2009): composite index of how comprehensively a Science Standard covers evolution.
  - Based on: mention of 'evolution'; coverage of biological, human, geological, cosmological evolution; their integration; and absence of creationist jargon/disclaimers.
- Rescaled 0 to 1: 0 = no/creationist coverage, 1 = very comprehensive coverage.
- Available for every state in 2000 and 2009, with the reform year in between.
- Treatment = the score in force when a student ENTERS high school (where evolution is mostly taught).
- Largest decreases: CT, LA, TX. Largest increases: KS, MS, FL.

Validating the Measure: Text Analysis of Science Standards

- Concern: does the evolution score really capture evolution coverage -- and ONLY evolution -- rather than broader science content?
- Collected the original text of 73 Science Standards (27 from 2000, 46 from 2009); OCR to machine-readable text.
- Built word-stem counts for the SAME 10 science topics used to classify NAEP questions (e.g. 'evolved' -> 'evolution', via Python regex).
- 10 topics: evolution, motion, matter/mass, energy, reproduction, climate change, pollution, the earth, tectonics, the universe.
  - Regress log word count on the evolution score, with state and year fixed effects (eq. 3); SEs clustered by state.
- Placebo logic: if reforms are evolution-specific, only the evolution word count should move with the score.

Text Analysis: Reforms Move Evolution Words, Not Other Science

Table A.XVI
Text Analysis: Effect of Evolution Coverage in Science Standards (as Measured by the Evolution Score) on Counts of Different Scientific Words in Science Standards

	Word Count of:	Counts of the following Non-Evolution Scientific Words:										Counts of all Words:
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Evolution	Motion	Matter and Mass	Energy	Reproduction	Climate Change	Pollution	Earth	Tectonics	Universe	Sum	Sum
Evolution Score	1.986***	0.505	1.137	0.725	0.494	0.933	0.357	1.186	1.195	1.202	0.798	0.300
	(0.736)	(1.061)	(1.031)	(1.022)	(0.665)	(0.745)	(1.863)	(1.566)	(1.294)	(1.801)	(1.078)	(1.431)
State FEs	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FEs	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
Mean of Dep. Var.	2.38	3.09	3.64	4.02	2.29	0.33	0.93	3.99	1.07	2.62	5.31	9.50
Std. Dev. of Dep. Var.	0.99	0.88	0.90	0.93	0.81	0.52	0.91	0.97	0.76	0.91	0.90	0.93
Adj. R-squared	0.359	0.387	0.403	0.321	0.362	0.278	0.146	0.474	0.157	0.307	0.400	0.517
Observations	73	73	73	73	73	73	73	73	73	73	73	73

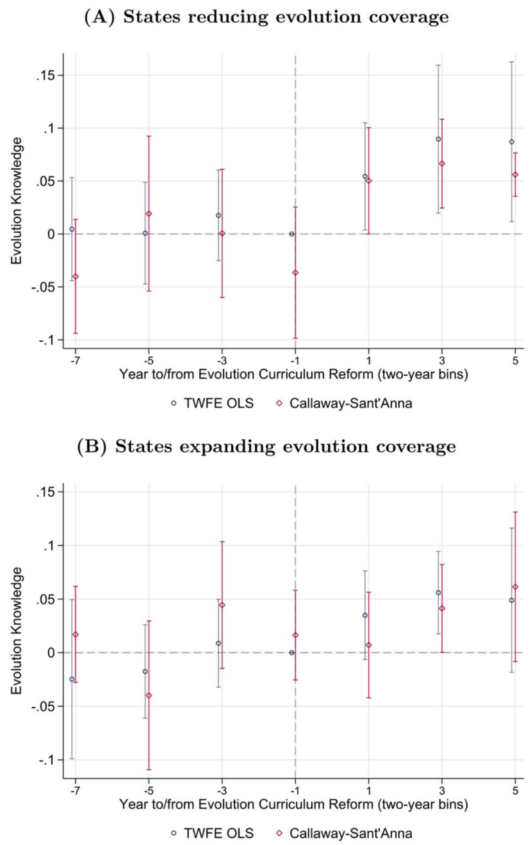
*Note: Online Appendix Table A.XVI (PDF p.92). Log word counts of 10 science areas (cols 1-10), the non-evolution sum (col 11), and all words (col 12) regressed on the evolution score with state + year FE. Only evolution (col 1) rises significantly (e^1.986 = 7.3x for score 1 vs 0, p=0.010, ***); every non-evolution coefficient is insignificant (col 11 p=0.463; col 12 p=0.835). Reforms shift evolution coverage specifically, not overall science content -- a placebo/validation of the treatment.*

Empirical Strategy: Staggered Reforms, Two-Way FE

- Baseline: Y = b * EvolutionScore + controls + state FE + cohort FE + survey-year FE (eq. 1); SEs clustered by state.
- b compares full evolution coverage (score=1) to no/creationist coverage (score=0), identified from adjacent cohorts around sharp reforms.
  - State FE absorb fixed state differences; cohort FE absorb national trends; the design nets out state shocks that hit adjacent cohorts equally.
- Robustness: state-specific linear/quadratic trends; binary treatment; balance tests of the score on predetermined covariates; dropping top/bottom 20% of the score.
- Callaway-Sant'Anna (2021) estimator addresses bias from staggered timing / heterogeneous effects.
- Event studies (eq. 2) run SEPARATELY for reducing vs. expanding states, since opposing reforms would otherwise cancel.

Result 1: Evolution Coverage Raises Evolution Knowledge

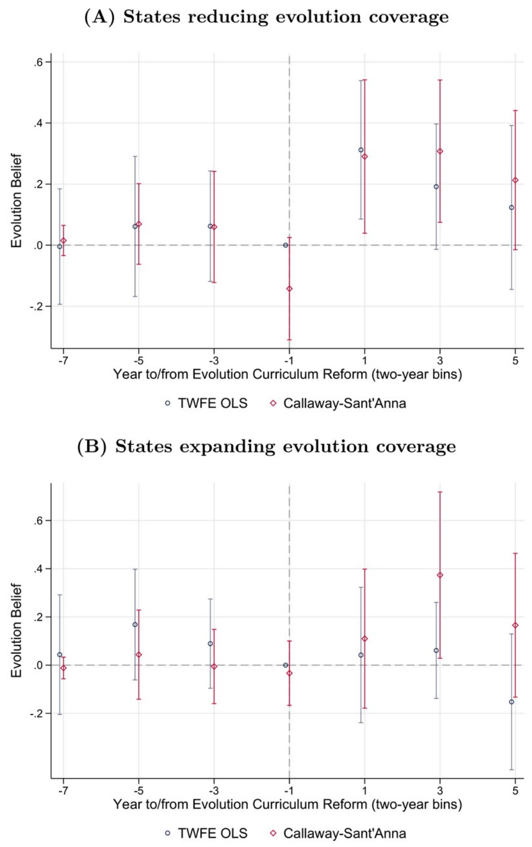
Figure II
Event-Study Graphs for Effect of Evolution Coverage in Science Standards on Evolution Knowledge in School



Note: Figure II (NAEP, end of high school). Moving from score 0 to 1 raises the share of evolution questions answered correctly by 6.5pp (p=0.005) -- 20% of the 32% mean. Event studies: flat, insignificant pre-trends (Panel A joint F=0.27, p=0.844); immediate, stable, significant post-reform jumps. Panel A = reducing states (outcome inverted for comparability), Panel B = expanding states; effects larger for reductions. TWFE (blue) and Callaway-Sant'Anna (red) agree.

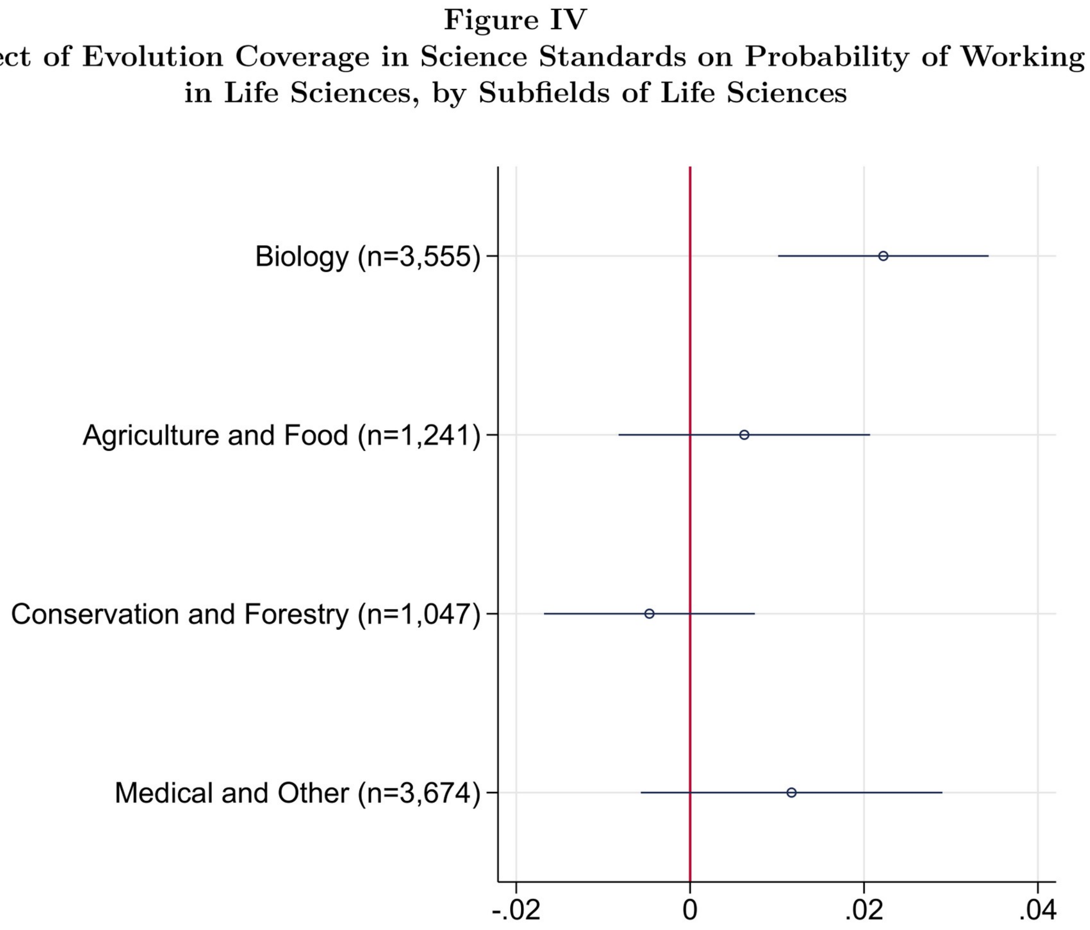
Result 2: Lasting Evolution Belief, No Crowd-Out

Figure III
Event-Study Graphs for Effect of Evolution Coverage in Science Standards on Evolution Belief in Adulthood



Note: Figure III (GSS, adulthood). Full vs. no coverage raises belief in evolution by 33.3pp -- 57% of the mean, a 79% persuasion rate; largest for Mainline Protestants. NO effect on religiosity or political attitudes, and no effect on non-evolution science attitudes (Tables A.VII-A.X) -- effects are neatly tied to evolution. Effects persist long after high school. Panel A = reducing states, Panel B = expanding states.

Result 3: Evolution Coverage Raises Working in Life Sciences



Note: Figure IV (ACS, adulthood). Full vs. no coverage raises the probability of working in life sciences by 23% of the sample mean. The effect is concentrated in BIOLOGY -- the subfield where evolution is taught (significant, CI above zero) -- while agriculture, conservation, and medical/other subfields show no effect. Science education can attract future STEM workers, with wider effects on innovation and growth.

Takeaways

- The CONTENT of science education has causal, lasting effects: scientific beliefs can be 'taught' in school and persist into adulthood.
- Expanded evolution coverage raises evolution knowledge, evolution belief, and entry into life-science careers -- without crowding out religiosity or shifting politics.
- The evolution score is a validated, evolution-specific treatment: text analysis shows reforms move evolution words 7x but leave other science content unchanged.
- Effects are consistently larger in absolute terms for reforms that REDUCE coverage.
- Policy: science curricula are a lever policymakers directly control to mitigate anti-scientific attitudes, with payoffs for public health, the environment, and the economy.